Luddy Undergraduate Research: Fall 2024
Student-Mentor Research Agreement

Scholar name: **Lucas Gottdenker**
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Mentor (PI) name: **Mitja Hmeljak**
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Co-Mentor (if any):
Co-Mentor email:

Credit hours: (default: 1)

> **Notes:** At IU Bloomington, a credit hour equates to 1 hour of instruction and 2 hours of student work per credit hour for approximately 15 weeks (BL-ACA-H13), constituting a total of 45 hours. In the context of mentored Undergraduate Research, we combine these for a total of 45 hours of student work per credit hour of research. Scaling this over 12 weeks (start of Week 3 to end of Dead Week), students and mentors should match their research expectations to 3.5-4 hours of work per week per credit hour, *including student research time, mentoring, and lab meetings.*
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> Mentors and students are encouraged to choose a "bite-sized," tangible advance towards an ongoing research project. Moreover, to encourage responsible research conduct, we recommend research objectives that *do not rely upon a specific research outcome* (e.g., prove that the Earth is an icosahedron), but rather emphasize performing an investigation or following a well-defined plan (e.g., evaluate seismography data with new analysis techniques to determine earth's shape).

Overall title (or theme):
Music Visualization

Specific research objectives / work for Fall 2024:
First and foremost, I will need to familiarize myself with the Unity game engine, as well as the C# programming language, in order to effectively use this platform to build a music visualization demo. I will also need to figure out what type of visualization technique I would like to replicate and determine how said technique can be translated into code. Finally, I will need to build a working demo and figure out what meaningful information can be gleaned from its representation of music (i.e. articulate why it is helpful/interesting).

Metrics for success (How will we decide if the student was successful?):
I would consider my work successful if, by the end of the semester, I am able to produce some kind of working music visualization demo in Unity. Considering that I am currently unfamiliar with Unity and C# (and quite new to programming in general), I think it is important that I keep my expectations at a reasonable level. Ideally, I would like to create a visualization project involving either self-similarity grids or isochords, as I find both of these visualization methods to be very interesting. I think that creating a working demo involving one of these visualization methods would be a reasonably attainable goal for the end of the semester.